The Sum Is Less Than Its Parts: Decomposing the Attention Write for RAG Source Attribution

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Abstract

In RAG workflows, it is often desirable to identify which passages from the source document contributed to generating the response. Existing approaches rely on textual similarity, attention weights, or repeated counterfactual inference, none of which observe the direct contribution of each source to the model's residual stream. We introduce GROUNDHOG, a training-free method that decomposes each attention head's output into source-specific writes and attributes their contributions to the answer's logits. Across three model families, two-sided GROUNDHOG achieves the highest average precision among evaluated baselines in RAG QA source attribution. We additionally show that the same source-partitioned representation supports GROUNDHOG$_S$, a sparse supervised extension for contextual hallucination detection. These results establish source-specific attention writes as a useful mechanistic basis for RAG source attribution.

1 Introduction

Retrieval-augmented generation (RAG) conditions a language model's responses on external sources to enhance the model's knowledge (Lewis et al., 2020). However, determining which part of the external source was used to generate a response remains challenging. A source document may be semantically similar but unrelated to the final response, or evidence for a claim may be distributed across several sources. Identifying those sources is important for attaching citations, inspecting retrieval failures, and understanding how a model combines evidence. Existing approaches infer source importance using lexical or semantic retrieval, entailment models, counterfactual source removal, or signals extracted from the generator itself (Gao et al., 2023; Cohen-Wang et al., 2024a; Qi et al., 2024a). Attention-based methods suggest a particularly simple approach: inspect how generated tokens attend to each source.

Attention-based attribution offers a direct view of how the model integrates context, and several methods use attention patterns to estimate source influence (Ferrando et al., 2022; Cohen-Wang et al., 2025). Attention weights are an intuitive measure of context contribution; however, they are just one component in an attention head. Source contribution also depends on the attention head's transformed value vector and its projection into the residual stream (Jain and Wallace, 2019a; Kobayashi et al., 2020): these steps remain comparatively unexplored in examining source usage. We inspect all of the attention head components together to determine not merely how much attention a source receives, but what that source contributes to the core residual stream.

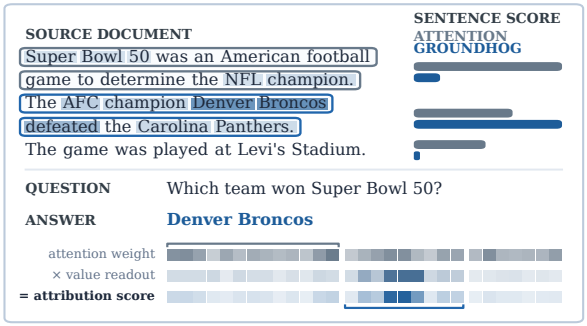


Figure 1: **GROUNDHOG answer attribution.** Attention (gray) is spread across the whole passage and is highest on the wrong sentence. GROUNDHOG (blue) weights each source word's attention by what that word writes toward the answer, which concentrates the score on the supporting sentence. Words are shaded by their GROUNDHOG score.

We introduce GROUNDHOG, a training-free method that decomposes each attention-head output into source-specific writes. Each component retains its attention weights, transformed values, and output projection, and the components sum exactly to the original attention write. GROUNDHOG reads each source vector in the direction of the answer token's unembedding and aggregates these scores across the answer, obtaining attributions for every source from a single forward pass. We evaluate this method over multiple contextual question answering datasets and demonstrate that it outperforms other source attribution methods. Across three models and datasets, GROUNDHOG obtains top precision@1 and average precision. Finally, a sparse supervised extension, GROUNDHOG$_S$, shows that the same source-resolved representation also supports contextual hallucination detection by comparing the total

quantity of attention head output sourced from context vs. that produced from its own generated tokens.

2 Background and Related Work

RAG source attribution. Retrieval-augmented generation conditions a language model on external documents, but generation alone does not identify which retrieved evidence contributed to a particular answer. Source-attribution methods address this problem by assigning scores to documents, passages, sentences, or individual context tokens. A useful distinction is between *corroborative* attribution, which asks whether a source logically supports an output, and *contributive* attribution, which asks whether the source affected the model's computation (Worledge et al., 2024). These objectives can agree, but neither entails the other: a passage may support an answer without having influenced its generation, while a passage may influence the model without constituting valid human-readable evidence.

External evidence matching. Corroborative methods compare the question or answer with candidate sources using lexical retrieval, dense representations, entailment models, or an LLM judge. Such methods are applicable to closed models and often align well with human notions of relevance, but their scores describe relationships among texts rather than the generating model's use of those texts. We therefore treat lexical similarity, dense similarity, NLI, and LLM-judgment methods as important external baselines rather than mechanistic explanations of the model's computation.

Counterfactual attribution. Perturbation methods estimate contributive influence by deleting or masking sources and measuring the resulting change in the answer distribution. Leave-one-out attribution provides a direct version of this test, while ContextCite learns a surrogate over source-removal interventions and AttriBoT introduces approximations that reduce the cost of leave-one-out attribution (Cohen-Wang et al., 2024b; Liu et al., 2025). TokenShapley instead estimates token-level influence through a Shapley-value construction (Xiao et al., 2025). These approaches can capture nonlinear changes in model behavior, but require repeated or perturbed inference and may conflate source removal with changes in prompt length, token positions, and attention normalization.

Model-internal attribution. White-box methods use representations already computed inside the model. MIRAGE combines internal activations and gradients to identify context-sensitive answer tokens and their influential source tokens (Qi et al., 2024b); HSSAVG localizes contextual evidence using hidden-state similarity (Phukan et al., 2024); and AttnUnionDep selects evidence from attention patterns augmented with dependency structure (Ding et al., 2025). These methods span gradients, hidden states, and attention, and can provide token-level explanations without training a separate external judge. Our evaluation includes representatives of these families alongside external and counterfactual baselines.

Attention and the residual write. Attention weights are an especially simple internal signal because they already associate a generated position with earlier context positions. However, an attention weight records where a head reads, not what that position contributes to the residual stream; interpreting attention alone as explanation can therefore be misleading (Jain and Wallace, 2019b). Hallucination detectors such as Lookback Lens compare attention mass assigned to retrieved context and generated history (Chuang et al., 2024), while TPA projects a head output toward the emitted token and apportions the result using attention mass (Lu et al., 2025). NoVo instead shows that total attention-head outputs contain hallucination-discriminative information, but does not preserve their source-specific addends (Ho et al., 2024). In contrast, GROUNDHOG partitions the attention-weighted value sum before the output projection is collapsed, retaining the exact vector written by each source and its additive contribution to the answer logit.

Hallucination signals. We treat contextual hallucination detection as a secondary application of the same source-partitioned representation. Prior detectors extract signals from hidden states, attention, feed-forward updates, or counterfactual context use. SAPLMA predicts truthfulness from hidden states (Azaria and Mitchell, 2023); ReDeEP characterizes competition between retrieved evidence and parametric knowledge in feed-forward and attention components (Sun et al., 2024); and LUMINA compares context and parametric-knowledge utilization using counterfactual inference (Yeh et al., 2025). These findings motivate GROUNDHOG$_S$, our supervised extension, but differ from the primary source-attribution problem: hallucination detection predicts whether an output is unsupported, whereas source attribution ranks the particular evidence associated with that output.

3 GROUNDHOG: Source Attribution from Attention Writes

Consider a language model that answers a query using retrieved sources $S_1, \ldots, S_K$. Model-internal attribution methods commonly use either attention weights or total attention-head outputs. Attention weights indicate which positions a head reads, but not the vector the head writes to the residual stream. The total head output describes the vector that is written, but not the positions from which it was obtained. GROUNDHOG combines the two: it decomposes each head's output over the attended positions and projects each component onto the logit of the answer token. The result is the direct contribution of each source to the answer.

3.1 Source-Specific Attention Writes

Let y_t be an answer token and p_t the position from which it is predicted. At layer ℓ and head h, let $\alpha_{t,j}^{\ell,h}$ denote the post-softmax attention weight from p_t to an earlier position j, and let $v_j^{\ell,h}$ be the value vector at j. The contribution of the head to the residual stream is

$$w_t^{\ell,h} = W_O^{\ell,h} \sum_{j<p_t} \alpha_{t,j}^{\ell,h} v_j^{\ell,h}, \tag{1}$$

where $W_O^{\ell,h}$ is the slice of the output projection corresponding to head h. The preceding positions are partitioned into disjoint groups $\mathscr{P}_t = \{S_0, S_1, \ldots, S_K\}$, where $S_1, \ldots, S_K$ are the retrieved source spans and S_0 contains all remaining positions, such as instructions, the question, and the answer prefix. The write attributable to group S_k is

$$w_{t,k}^{\ell,h} = W_O^{\ell,h} \sum_{j\in S_k} \alpha_{t,j}^{\ell,h} v_j^{\ell,h}. \tag{2}$$

Since the partition is applied to the terms of the sum before they are added,

$$w_t^{\ell,h} = \sum_{k=0}^{K} w_{t,k}^{\ell,h}. \tag{3}$$

The value vector $v_j^{\ell,h}$ is computed from the residual stream at position j. In a causal model, this residual contains information from the token at j and from all preceding positions. The components in Equation 2 are therefore attributed to positions, that is, to the residual states read by the head, and not to the identities of individual input tokens. The decomposition is exact for this quantity and requires no ablation or re-encoding. For a source span, the attribution is to the states at the positions of that span, which are the states from which the head reads the source.

3.2 Answer-Conditioned Logit Attribution

To measure whether a source write supports the realized answer, each write is projected onto the logit direction of the corresponding answer token. Let x_t be the final residual at position p_t, U_{y_t} the unembedding vector of y_t, and γ the learned gain of the final RMSNorm. For a given forward pass,

$$\text{logit}_{y_t} = U_{y_t}^\top \text{RMSNorm}(x_t) = u_{y_t}^\top x_t, \qquad u_{y_t} = \frac{\gamma \odot U_{y_t}}{\rho_t}, \tag{4}$$

where $\rho_t = \sqrt{d^{-1}\|x_t\|_2^2 + \varepsilon}$ is fixed by that forward pass. Because the residual stream is a sum of component writes, the contribution of source S_k to the attention-derived portion of the logit is

$$q_{t,k} = \sum_{\ell,h} u_{y_t}^\top w_{t,k}^{\ell,h}. \tag{5}$$

At the level of individual positions, the contribution of position j is

$$q_{t,j} = \sum_{\ell,h} \alpha_{t,j}^{\ell,h} \left(u_{y_t}^\top W_O^{\ell,h} v_j^{\ell,h} \right). \tag{6}$$

	token share (%)		
	base rate	attention	GROUNDHOG
Content	59.2	62.7	**86.9**
Function words	26.3	17.7	**7.0**
Punctuation, symbols	14.5	19.6	**6.1**

Table 1: **Token-class distributions.** Attention roughly follows the token baseline, whereas GROUNDHOG concentrates on content tokens.

A position receives a high score only when the head attends to it and the value written from it is aligned with the answer token. Attention weight alone does not determine the score.

The quantity $q_{t,k}$ is the direct contribution of the attention writes from S_k to the answer logit. It does not include effects that are mediated by later layers, for example when a source write changes the attention pattern of a later head or the input to a feed-forward layer. It also covers only the attention component of the logit; the remaining portion comes from the token embedding and the feed-forward writes.

Scores are aggregated over the answer tokens and over the positions belonging to a source:

$$\text{GH}(S_k \mid y) = \sum_t q_{t,k} = \sum_t \sum_{j\in S_k} q_{t,j}. \tag{7}$$

The resulting signed score is used to rank retrieved passages, sentences, or other source spans by their direct contribution to the answer. The computation requires one forward pass and requires no additional training.

4 GROUNDHOG: Source Attribution from Attention Writes

Consider a language model answering a query from retrieved sources $S_1, \ldots, S_K$. Existing model-internal attribution methods commonly inspect attention weights or total attention-head outputs. Attention weights describe where a head reads, but omit the value vectors and output projection that determine what the head writes to the residual stream. Conversely, the total head output preserves the write but no longer identifies which source positions produced it. GROUNDHOG retains both: it decomposes the realized attention write exactly by source and measures each source-specific write's contribution to the answer logits.

4.1 Exact Source-Specific Writes

Let y_t be an answer token and p_t the causal position from which it is predicted. At layer ℓ and head h, let $\alpha_{t,j}^{\ell,h}$ denote the post-softmax attention weight from p_t to an earlier position j, and let $v_j^{\ell,h}$ be its value vector. The head's contribution to the residual stream is

$$w_t^{\ell,h} = W_O^{\ell,h} \sum_{j<p_t} \alpha_{t,j}^{\ell,h} v_j^{\ell,h}, \tag{8}$$

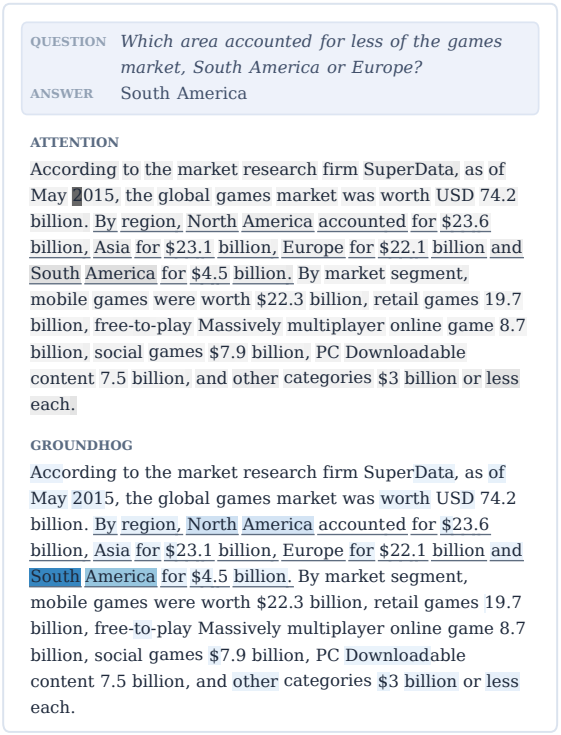


Figure 2: **Source attribution comparison.** Attention attribution is more diffuse and noisy; GROUNDHOG tends to produce more localized and relevant attributions.

where $W_O^{\ell,h}$ is the head's slice of the output projection. Partition the preceding positions into disjoint groups $\mathscr{P}_t = \{S_0, S_1, \ldots, S_K\}$, where $S_1, \ldots, S_K$ are retrieved source spans and S_0 contains all remaining positions, such as instructions, the question, and the answer prefix. The write attributable to group S_k is

$$w_{t,k}^{\ell,h} = W_O^{\ell,h} \sum_{j \in S_k} \alpha_{t,j}^{\ell,h} v_j^{\ell,h}. \tag{9}$$

Because this partitions the addends before they are summed,

$$w_t^{\ell,h} = \sum_{k=0}^{K} w_{t,k}^{\ell,h}. \tag{10}$$

Thus, source partitioning introduces no approximation and requires no ablation or re-encoding. Unlike a post hoc allocation based only on attention mass, it preserves the vector actually written by each source into the residual stream.

4.2 Answer-Conditioned Logit Attribution

To determine whether a source write supports the realized answer, we project it onto the corresponding answer-token logit direction. Let x_t be the final residual at position p_t, U_{y_t} the unembedding vector of y_t, and γ the learned gain of the final RMSNorm. For the realized forward pass,

$$\text{logit}_{y_t} = U_{y_t}^\top \text{RMSNorm}(x_t) = u_{y_t}^\top x_t, \qquad u_{y_t} = \frac{\gamma \odot U_{y_t}}{\rho_t}, \tag{11}$$

where $\rho_t = \sqrt{d^{-1}\|x_t\|_2^2 + \varepsilon}$ is fixed by that forward pass. Since the residual stream is additive, the contribution of source S_k to the attention-derived portion of this logit is

$$q_{t,k} = \sum_{\ell,h} u_{y_t}^\top w_{t,k}^{\ell,h}. \tag{12}$$

Equivalently, before pooling into source spans, the contribution of individual source token j is

$$q_{t,j} = \sum_{\ell,h} \alpha_{t,j}^{\ell,h} \left(u_{y_t}^\top W_O^{\ell,h} v_j^{\ell,h} \right). \tag{13}$$

This expression exposes the distinction from raw attention: a position receives high attribution only when the model both attends to it and writes a value that supports the realized answer token. We aggregate over the answer and over the tokens belonging to a source:

$$\text{GH}(S_k \mid y) = \sum_t q_{t,k} = \sum_t \sum_{j \in S_k} q_{t,j}. \tag{14}$$

The resulting signed score ranks retrieved passages, sentences, or arbitrary source spans according to their direct contribution to the answer. GROUNDHOG requires one instrumented forward pass, no fitted parameters, and no counterfactual source deletion.

5 Source Attribution Evaluation

Benchmarks. We evaluate whether attribution scores recover human-annotated evidence in both multi-hop and single-hop question answering. For multi-hop evaluation, we use 500 examples from the HotpotQA distractor validation set, treating each context sentence as a candidate source and the annotated supporting facts as positives, and 500 examples from the MuSiQue-Answerable development set, treating its 20 paragraphs as candidate sources and the annotated supporting paragraphs as positives. For single-hop evaluation, we use all 12 MRQA subsets, with 100 examples per subset; candidate sources are context sentences, and a sentence is positive when it contains the annotated extractive answer span. We additionally use 224 eligible questions from the official WikiQA test set, for which candidate answer sentences have human relevance labels. This combination tests recovery of complementary evidence chains, localization of answer-bearing text across diverse reading-comprehension domains, and direct ranking of human-judged answer sentences.

Models and attribution target. We evaluate Llama-2-7B-Chat, Llama-3.1-8B-Instruct, and Qwen3-4B-Instruct-2507. Every method receives the same ordered source set, question, and gold answer. Model-internal

methods attribute a teacher-forced forward pass over that answer, which isolates source attribution from variation in generation accuracy. For WikiQA, the target is a separately annotated answer phrase rather than a positive candidate sentence. External answer-matching baselines receive the concatenated question and answer, as indicated by $q{+}a$ in Table 2.

Baselines. We compare against controls, external evidence matching, model-internal attribution, and counterfactual attribution. Random ranking and source length test chance performance and length bias. BM25 and BGE score lexical and dense similarity to the question and answer, while DeBERTa-NLI scores whether each source entails them. The internal baselines are summed attention, input-gradient saliency, MIRAGE-EX, and AttnTrace. We report both AttnTrace-1, its single-forward attention map and pooling rule, and full AttnTrace, which averages across 30 context subsamples. The counterfactual baselines are ARC-JSD, which measures the predictive-distribution change after deleting each source; exact leave-one-out, which measures the change in answer log likelihood; and ContextCite-32, fit from 32 source-removal masks. Thus, the comparison includes inexpensive retrieval and single-pass methods as well as substantially more expensive perturbation methods.

Metrics and significance. We report precision at one (P@1), which tests whether the top-ranked source is relevant, and average precision (AP), which evaluates the complete ranking while emphasizing relevant sources near the top. AP is the primary metric for HotpotQA and MuSiQue because each question requires multiple supporting sources. MRQA results macro-average P@1 and AP over its 12 datasets so that large subsets do not dominate. We test the AP improvement over the strongest non-GROUNDHOG method using a two-sided paired randomization test at the question level and apply Holm correction within each benchmark; positive corrected differences are marked by $^{*}p < 0.05$, $^{**}p < 0.01$, and $^{***}p < 0.001$.

Faithfulness. Agreement with human evidence annotations measures whether a method identifies sources that support the answer, but not whether the model's output depends on those sources. We therefore also evaluate faithfulness using the linear datamodeling score (LDS; **?**Cohen-Wang et al., 2024a). For each question, we sample 64 ablation masks, each of which removes every source independently with probability 0.5, and compute the log-probability of the gold answer under each ablated context. The LDS is the Spearman correlation between these log-probabilities and the sum of the attribution scores of the retained sources, averaged over questions. ContextCite is fit on 32 separate masks that are disjoint from the evaluation masks. Because it is fit to predict the same quantity that the LDS measures, we report it as a fitted reference rather than as a directly comparable baseline. Faithfulness is evaluated on 300 extractive examples from HotpotQA and 300 examples from MRQA, with 25 examples from each of the 12 subsets.

6 Source Attribution Results

Two-sided attribution recovers multi-hop evidence. Table 2(a) shows that two-sided GROUNDHOG obtains the highest AP for all three models on MuSiQue and for Llama-3.1 and Qwen3 on HotpotQA. Its improvement over the strongest non-GROUNDHOG baseline is significant in each of these five settings. On HotpotQA with Llama-2, it remains second by only 0.010 AP to BGE. Relative to one-pass GROUNDHOG, the two-sided ranking raises AP from .613–.667 to .742–.802 on HotpotQA and from .630–.681 to .753–.819 on MuSiQue. P@1 is unchanged by construction because the answer-conditioned source remains the first-ranked anchor; the gain comes from recovering additional supporting sources through question-conditioned attribution. This pattern supports the intended division of labor: the answer side identifies evidence that directly writes the answer, whereas the question side retrieves evidence that establishes an intermediate entity or relation.

One-pass attribution is strongest for single-hop localization. Table 2(b) shows that ordinary one-pass GROUNDHOG achieves the best P@1 and AP for all three models on the 12-dataset MRQA macro-average. Its AP ranges from .801 to .803, compared with .702, .781, and .702 for the respective strongest baselines, and its P@1 ranges from .706 to .713. On WikiQA, GROUNDHOG is strongest in five of the six model–metric comparisons. The only exception is Llama-3.1 AP, where .9695 is 0.0008 below AttnTrace-1; GROUNDHOG nevertheless attains the highest P@1 for that model and the highest P@1 and AP for both other models. The consistent single-hop result is notable because GROUNDHOG uses neither a retriever trained for semantic similarity nor repeated source deletions: it reads the contribution already written during one teacher-forced model pass.

Faithfulness. Leave-one-out achieves a higher LDS than GROUNDHOG (.644 vs. .552), but it requires one additional forward pass for each source, with a median of 36 sources per question. GROUNDHOG achieves a higher LDS than ARC-JSD, which has the same cost as leave-one-out, by .153 (95% CI [.103, .207]). ContextCite, which is fit on 32 separate ablation masks, achieves .681. For 11 of the 100 questions, all of its fitted coefficients are zero and its LDS is undefined; the results in Table **??** are therefore computed on the remaining 89 questions for all methods.

Table **??** reports the LDS for each method. GROUNDHOG achieves a higher LDS than attention and BGE on both datasets (HotpotQA: .xxx vs. .xxx and .xxx; MRQA: .xxx vs. .xxx and .xxx). The difference from attention, which corresponds to GROUND-

(a) Human supporting-evidence recovery

Method	HotpotQA			MuSiQue		
	Llama-2	Llama-3.1	Qwen3	Llama-2	Llama-3.1	Qwen3
	P@1 / AP					
Controls and retrieval						
Random	.086/.154	.086/.154	.086/.154	.148/.257	.148/.257	.148/.257
Source length	.112/.199	.110/.198	.108/.200	.176/.289	.178/.288	.186/.290
BM25 (q+a)†	**.814**/.696	.814/.696	.814/.696	.794/.673	.796/.673	.796/.673
BGE (q+a)†	.792/**.752**	.792/.752	.792/.752	.790/.711	.794/.713	.794/.713
DeBERTa-NLI (q+a)†	.440/.372	.440/.372	.440/.372	.378/.390	.374/.389	.374/.389
Context-attribution baselines						
Attention sum	.652/.533	.852/.728	.610/.536	.562/.488	.920/.699	.758/.579
Input-gradient saliency	.376/.463	.174/.368	.288/.419	.512/.526	.280/.466	.396/.495
MIRAGE-EX	.324/.276	.458/.331	.492/.353	.476/.385	.544/.400	.570/.426
AttnTrace-1	.590/.497	.856/.739	.334/.435	.682/.528	.938/.748	.288/.456
AttnTrace	.488/.468	.798/.696	.480/.448	.696/.543	.932/.729	.438/.445
ARC-JSD	.788/.649	**.868**/.741	**.856**/.710	.792/.645	.912/.759	.904/.748
Exact leave-one-out	.772/.528	.674/.452	.728/.519	.858/.575	.916/.602	.940/.663
ContextCite-32	.728/.480	.722/.489	.764/.547	**.928**/.591	.946/.619	**.956**/.664
Our methods						
GROUNDHOG	.796/.613	.856/.667	.834/.627	.902/.630	**.954**/.681	.944/.650
GROUNDHOG-two-sided	.796/.742	.856/**.802*****	.834/**.782***	.902/**.753*****	**.954/.813*****	.944/**.819*****

(b) Single-hop source localization

Method	MRQA (12-dataset macro)			WikiQA		
	Llama-2	Llama-3.1	Qwen3	Llama-2	Llama-3.1	Qwen3
	P@1 / AP					
Controls and retrieval						
Random	.118/.283	.118/.282	.118/.282	.172/.362	.172/.362	.172/.362
Source length	.199/.380	.197/.379	.197/.379	.271/.469	.281/.474	.281/.479
BM25 (q+a)†	.577/.702	.577/.702	.577/.702	.907/.930	.907/.930	.907/.930
BGE (q+a)†	.543/.678	.544/.677	.544/.677	.914/.936	.914/.936	.914/.936
DeBERTa-NLI (q+a)†	.462/.582	.460/.580	.460/.580	.778/.841	.778/.841	.778/.841
Context-attribution baselines						
Attention sum	.534/.665	.669/.761	.600/.711	.792/.867	.955/.964	.923/.944
Input-gradient saliency	.226/.441	.189/.407	.206/.432	.527/.685	.448/.644	.473/.667
MIRAGE-EX	.400/.528	.441/.551	.514/.603	.710/.789	.765/.830	.821/.861
AttnTrace-1	.461/.635	.690/.781	.572/.693	.615/.788	.973/**.9703**	.860/.910
AttnTrace	.388/.571	.603/.728	.450/.619	.525/.702	.860/.893	.665/.780
ARC-JSD	.599/.690	.635/.715	.618/.697	.923/.929	.964/.965	.941/.947
Exact leave-one-out	.612/.683	.553/.625	.577/.642	.937/.918	.900/.884	.900/.886
ContextCite-32	.623/.701	.573/.657	.602/.683	.937/.926	.891/.890	.914/.901
Our method						
GROUNDHOG	**.706/.801**	**.709/.802**	**.713/.803**	**.973/.967**	**.977**/.9695	**.977/.972**

Table 2: Source localization across multi-hop and single-hop QA. Each entry reports P@1/AP. Panel (a) recovers human-annotated supporting evidence in multi-hop QA; AP is primary. Panel (b) localizes human-annotated answer evidence, macro-averaging each metric over the 12 MRQA subsets. Bold and underline mark the best and second-best displayed result for each metric and model within a benchmark. Stars mark a positive paired AP improvement over the strongest non-GROUNDHOG method after Holm correction ($^{*}p < 0.05$, $^{**}p < 0.01$, $^{***}p < 0.001$). Model-internal methods explain the gold answer; † external retrieval, entailment, and answer-matching baselines additionally receive that answer.

HOG without the value vectors and output projection, is +.xxx on HotpotQA and +.xxx on MRQA. Comparison with leave-one-out and ARC-JSD, and with ContextCite as a fitted reference. On HotpotQA, BGE achieves higher AP against the human evidence annotations (Appendix **??**), but its LDS is .xxx lower than that of GROUNDHOG. This indicates that the sources annotated as supporting evidence are not always the sources on which the model's output depends.

Answer and question conditioning serve different evidence structures. The contrast between panels clarifies when the second attribution side is useful. In MRQA and WikiQA, the answer-bearing source is itself the target, so direct answer-conditioned attribution is sufficient. In HotpotQA and MuSiQue, one supporting source may write the final answer while another supplies a bridge expressed more directly in the question; answer-only attribution consequently finds the anchor but can under-rank the bridge. Two-sided GROUNDHOG addresses this failure without fitting a fusion model or performing source ablations.

7 Extension to Contextual Hallucination Detection

Having evaluated source-specific writes directly against human evidence annotations, we next ask whether the same decomposition supports a related but distinct task: determining whether a generated response is grounded in its context. Source attribution ranks the particular sources associated with an answer, whereas contextual hallucination detection predicts whether sufficient support exists at all. The decomposition in Equation 9 accommodates both problems simply by changing the partition of preceding positions. We group the prompt and retrieved document into a context set C and previously generated tokens into a response-history set R, then use the resulting provenance-specific writes as training-free and supervised grounding signals.

7.1 Context and Response-History Writes

Following prior model-internal detectors, we partition preceding positions into a context set C, containing the prompt and retrieved document, and a response-history set R, containing previously generated tokens. Equation 9 then gives

$$w^{\ell,h}_{c,t} = W^{\ell,h}_O \sum_{j\in C} \alpha^{\ell,h}_{t,j} v^{\ell,h}_j, \qquad w^{\ell,h}_{r,t} = W^{\ell,h}_O \sum_{j\in R} \alpha^{\ell,h}_{t,j} v^{\ell,h}_j. \tag{15}$$

These are exact complements within the attention write: $w^{\ell,h}_t = w^{\ell,h}_{c,t} + w^{\ell,h}_{r,t}$. We additionally retain the feed-forward update $w^{\ell}_{f,t}$ as a non-attention channel through which parametric information may enter.

Figure **??** shows that this partition is informative: larger context writes are generally associated with supported tokens, whereas larger response-history writes are associated with hallucination. Their opposing associations become weaker in the total, unpartitioned write. Projecting the context write onto the emitted-token direction recovers the token-level GROUNDHOG score for the context as a whole:

$$s_c(t) = \sum_{\ell,h} u^{\top}_{y_t} w^{\ell,h}_{c,t}. \tag{16}$$

This provides a training-free diagnostic of contextual support. In contrast, the analogous directional projection of the response-history write is weak, although the response write itself remains predictive. This motivates a learned extension that combines distributed response-history signals.

7.2 GROUNDHOG$_S$: A Supervised Detector

For every head, GROUNDHOG$_S$ extracts three features:

$$m^h_t = \|w^h_{r,t}\|, \qquad c^h_t = \cos(w^h_{r,t}, w^h_{c,t}), \qquad f^h_t = \cos(w^h_{r,t}, w^{\ell}_{f,t}). \tag{17}$$

For response-level detection, we mean-pool each feature over the response positions:

$$\bar{m}^h = \frac{1}{|r|}\sum_{t\in r} m^h_t, \qquad \bar{c}^h = \frac{1}{|r|}\sum_{t\in r} c^h_t, \qquad \bar{f}^h = \frac{1}{|r|}\sum_{t\in r} f^h_t. \tag{18}$$

We standardize features using training-split statistics and fit a separate regularized logistic regression to each feature family. The final hallucination score sums their pre-sigmoid outputs:

$$S_{\text{GROUNDHOG}_S} = z_m(\bar{m}) + z_c(\bar{c}) + z_f(\bar{f}). \tag{19}$$

We use an ℓ_1 penalty to obtain sparse head selection; tuned ℓ_2 regularization performs similarly. Unlike GROUNDHOG, GROUNDHOG$_S$ requires labeled hallucination data and is evaluated as a supervised extension rather than as the paper's primary source-attribution method.

8 Hallucination Detection Evaluation

9 Hallucination Detection Results

10 Analysis and Discussion

11 Conclusion

We introduced GROUNDHOG, a training-free method for attributing language-model answers to retrieved sources by decomposing each attention head's residual-stream write into its exact source-specific addends. Unlike raw attention, GROUNDHOG retains both where a head reads and what the attended values write toward the realized answer; unlike counterfactual attribution, it requires no source deletion or repeated inference. Across three model families, two-sided GROUNDHOG

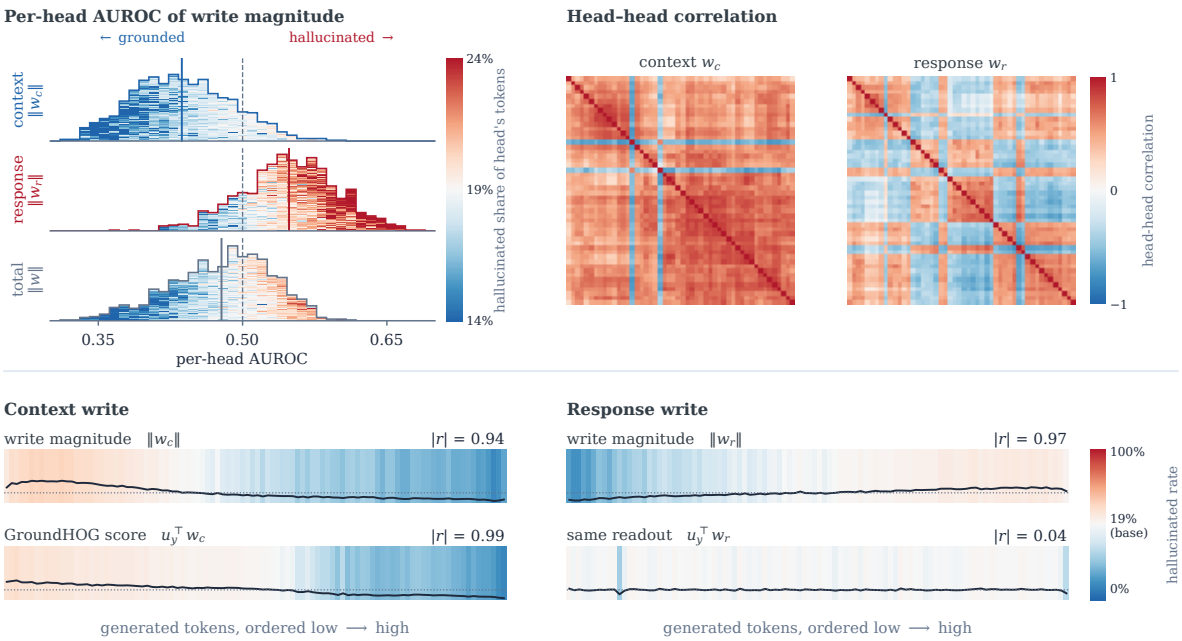


Figure 3: Caption

Method	RAGTruth			RAGognize (answerable)		
	L2-7B	M-7B	Q3-4B	L2-7B	M-7B	Q3-4B
ReDeEP	0.782/0.812	0.848/0.837	0.802/0.821	0.753/0.694	0.819/0.726	0.852/0.881
TPA	–	–	–	–	–	–
SAPLMA	0.663/0.625	0.738/0.729	0.653/0.641	0.669/0.557	0.739/0.592	0.701/0.718
NoVo (voting)	0.781/0.755	0.801/0.810	0.711/0.746	0.705/0.629	0.601/0.433	0.724/0.761
NoVo†	0.796/0.813	0.853/0.850	0.820/0.843	0.776/0.737	0.862/0.777	0.912/0.931
Lookback Lens	0.710/0.714	0.755/0.739	0.741/0.733	0.689/0.619	0.839/0.748	0.896/0.922
Lookback Lens†	0.819/0.825	0.851/0.851	0.819/0.829	0.788/0.750	0.871/0.797	0.919/0.936
GROUNDHOG	**0.821/0.848**	**0.873/0.874***	**0.825/0.850**	**0.805/0.768***	**0.873/0.797**	**0.927/0.942***

Method	RAGBench			HDM-Bench (context)		
	L2-7B	M-7B	Q3-4B	L2-7B	M-7B	Q3-4B
ReDeEP	0.721/0.468	0.752/0.504	0.757/0.503	0.830/0.923	0.849/0.933	0.847/0.928
TPA	–	–	–	–	–	–
SAPLMA	0.656/0.333	0.666/0.341	0.691/0.348	0.816/0.922	0.794/0.907	0.812/0.910
NoVo (voting)	0.643/0.296	0.669/0.348	0.728/0.460	0.751/0.895	0.751/0.882	0.765/0.888
NoVo†	0.772/0.468	0.789/0.538	0.825/0.553	0.856/0.941	0.833/0.922	0.879/0.947
Lookback Lens	0.689/0.398	0.742/0.463	0.773/0.465	0.819/0.919	0.827/0.921	0.844/0.929
Lookback Lens†	0.775/0.514	**0.804/0.548**	0.839/0.596	0.863/0.939	0.870/0.943	0.886/0.949
GROUNDHOG	**0.795/0.528****	0.799/0.554	**0.854/0.616****	**0.873/0.946**	**0.872/0.941**	**0.894/0.951**

Table 3: **Self-generated (top) and proxy (bottom) benchmarks.** HDM-Bench uses the `is_ctx_hallucination` label, excluding common-knowledge cases; RAGognize is answerable only. HDM evaluated using 5-fold stratified CV, all other results are response-level AUROC/AUPRC on the offical test set. Best per column bold; */**/*** represents the significance of GROUNDHOG$_S$ vs. the next-best baseline at $p < 0.05/0.01/0.001$ (paired bootstrap).

achieves the highest AP in all three MuSiQue settings and two of three HotpotQA settings, while one-pass GROUNDHOG achieves the highest macro P@1 and AP across 12 MRQA datasets and the strongest result in five of six WikiQA model–metric settings. We further showed that the same decomposition can be repartitioned into context and response-history writes, providing training-free grounding information and features for the supervised GROUNDHOG$_S$ hallucination detector. Together, these results establish source-

specific attention writes as a simple mechanistic representation for tracing generated answers to their evidence and, more broadly, for measuring whether generation remains grounded in context.

Limitations

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A Transformer Block Components

A decoder-only transformer maintains a residual stream that is carried forward through the network. Two subcomponents sequentially apply additive writes to the residual: a multi-head self-attention sublayer and a feed-forward network (FFN). In the models we study both are pre-normalization blocks and the normalization is RMSNorm; we write $\mathrm{Norm}(\cdot)$ for it.

Attention heads. The full model consists of layers of transformer blocks, each containing an attention sublayer consisting of multiple attention heads. For simplicity of notation, we fix a single layer ℓ and a single token position i corresponding to the output token y_i, writing x for the residual x_i^ℓ being updated and x_j for the residual at an attended position $j \le i$, so that only the head index h is noted explicitly unless necessary. In a pre-normalization transformer, x is first normalized. Then, for head h, the normalized residual is projected into queries, keys, and values:

$$q^h = W_Q^h \mathrm{Norm}(x), \quad k_j^h = W_K^h \mathrm{Norm}(x_j), \quad v_j^h = W_V^h \mathrm{Norm}(x_j). \tag{20}$$

The attention weight assigned to a previous position $j \le i$ is

$$\alpha_j^h = \mathrm{softmax}_{j \le i}\left(\frac{{q^h}^\top k_j^h}{\sqrt{d_h}}\right). \tag{21}$$

The head output is first computed as a weighted sum of value vectors. The outputs from all heads are then concatenated and projected back into the residual stream:

$$z^h = \sum_{j \le i} \alpha_j^h v_j^h, \tag{22}$$

$$x = x + W_O\left[z^1; z^2; \ldots; z^H\right]. \tag{23}$$

Feed-forward network. After attention updates the residual stream, the residual is again normalized, and the FFN then applies a learned nonlinear transformation and writes another vector into the residual stream. The FFN acts on the whole residual rather than per head, so it carries no head index; we write x' for the updated residual passed to the next layer:

$$x' = x + \mathrm{FFN}\left(\mathrm{Norm}(x)\right). \tag{24}$$

B Classifier Regularization, Sparsity, and Hyperparameters

Protocol. Every supervised feature is standardized to zero mean and unit variance using training-fold statistics before the classifier is fit; this makes the ℓ_1 penalty scale-invariant so heads are selected by discriminative value, not by the raw scale of their writes (§**??**). We use `sklearn LogisticRegression` with `class_weight=balanced`; ℓ_1 uses the `liblinear` solver and ℓ_2 the `lbfgs` solver. The strength C is selected once on a coarse grid $C \in \{0.01, 0.03, 0.05, 0.1, 0.3, 1\}$ and held fixed across all datasets and readers (no per-cell tuning); GROUNDHOG uses $C = 0.06$.

Tuned ℓ_1 vs. tuned ℓ_2: sparsity is for interpretability, not accuracy. At matched, tuned regularization the two penalties are interchangeable: Table 4 reports each method under its best fixed C, and tuned ℓ_1 and tuned ℓ_2 differ by at most 0.003 for every method, including GROUNDHOG (0.851 either way). We retain ℓ_1 not for accuracy but because its sparsity yields the interpretable per-head selection the analysis relies on, and because it is robust across C (a broad plateau, 0.851 for $C \in [0.05, 0.07]$) whereas a tuned ℓ_2 peaks sharply. The head-budget curves (Fig. 4, detailed below) show the same: a properly-regularized ℓ_2 coincides with ℓ_1 at every budget, while the default $C = 1$ ℓ_2 declines as its weight spreads over redundant heads. (An earlier claim that dense ℓ_2 trails by up to 0.066 used that under-regularized $C = 1$ model.)

Method	ℓ_1 (best C)	ℓ_2 (best C)
GROUNDHOG	**0.8510** ($C = 0.06$)	**0.8510** ($C = 0.003$)
Lookback Lens	0.8419 ($C = 0.1$)	**0.8449** ($C = 0.003$)
NoVo	0.8402 ($C = 0.05$)	**0.8408** ($C = 0.001$)
ReDeEP	0.8031 ($C = 0.1$)	**0.8044** ($C = 0.03$)
TPA	**0.7555** ($C = 1$)	**0.7555** ($C = 1$)

Table 4: **Sensitivity to the logistic-regression penalty.** Mean AUROC across the 12 evaluation settings under the best fixed value of C for each method and penalty (GROUNDHOG and each steelmanned baseline read through the same standardized linear head). Tuned ℓ_1 and ℓ_2 regularization perform nearly identically, and GROUNDHOG remains strongest under either penalty. (For reference, the Lookback Lens classifier as originally published uses sklearn's default ℓ_2, C=1, unstandardized, which scores 0.777 here—so our steelman is strictly more generous than the published detector.)

B.1 How many heads are needed?

If grounding is read from a small set of heads (§**??**), how few suffice? For each of GROUNDHOG's three feature families—the response-write magnitude $\|w_r\|$ and the two agreement cosines $\cos(w_c, w_r)$, $\cos(w_r, w_f)$—we rank heads by the family's per-head

ℓ_1 weight and restrict it to the top-k heads only. To avoid feature-selection leakage the ranking is recomputed inside each training fold; Figure 4 reports the resulting AUROC against k for three model families.

Two things hold across all three architectures. First, each family is *individually predictive and sparse*: a handful of heads recovers most of its accuracy and it saturates by $k \approx 20$–50, so the grounding signal is linearly decodable from a small set of heads in *each* stream. Second, what matters is regularization *strength*, not the penalty form: a properly-regularized dense ℓ_2 (tuned C, dashed) coincides with the sparse ℓ_1 (solid) at every budget. An apparent ℓ_1 advantage appears only for an *under*-regularized ℓ_2 (the default C=1, not plotted), whose weight spreads over the redundant heads so that its AUROC *declines* as the budget grows past $k \approx 50$–75; tuning C removes the gap entirely. We nonetheless use ℓ_1, because its sparsity is robust across C and yields the interpretable per-head selection this analysis relies on (§**??**).

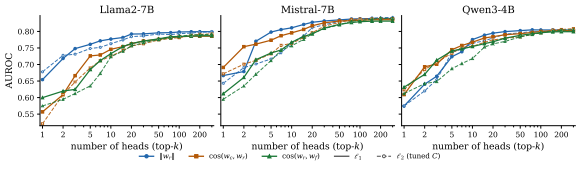


Figure 4: **Head budget.** AUROC of each of GROUNDHOG's three feature families ($\|w_r\|$, $\cos(w_c, w_r)$, $\cos(w_r, w_f)$) restricted to its top-k heads (ranked by per-head weight, leakage-free CV; RAGTruth), for three models, under ℓ_1 (solid) and tuned-C ℓ_2 (dashed). Across all models, every family is individually predictive and sparse (saturating by $k \approx 20$–50). The tuned ℓ_2 coincides with ℓ_1 at every budget—the penalty form does not matter, only its strength; an under-regularized C=1 ℓ_2 (not plotted) instead peaks around $k \approx 50$–75 then declines as its weight spreads over redundant heads.

C Component and Feature Ablations

The final detector reads three per-head feature families, $m = \|w_r\|$, $c = \cos(w_c, w_r)$, and $f = \cos(w_r, w_f)$ (§**??**). Table **??** evaluates each family alone and every combination, contrasts the per-family sum with a joint ℓ_1, and compares response- vs. context-anchoring; the pooling ablation compares response-token aggregations.

C.1 Token pooling

Table 6 compares mean (default), max, and last-token-only pooling of the per-head Gram.

Pooling over response tokens	AUROC	AUPRC
Mean (GROUNDHOG, default)	**0.799**	**0.863**
Max	0.772	0.845
Last output token only	0.742	0.821

Table 6: **Token-pooling ablation for GROUNDHOG** (RAGTruth, Llama2-7B, 5-fold CV). The per-head write magnitudes are aggregated over response tokens by mean (default), max, or by taking only the last output token (as last-token detectors such as NoVo and SAPLMA do). Mean-pooling is best by a clear margin: a hallucinated response often contains only a *local* unsupported span, and averaging the per-token evidence captures it, whereas the final token's writes mostly reflect its own generation and miss mid-response fabrications. This is why GROUNDHOG integrates over the whole response rather than reading a single token.

C.2 Combining GROUNDHOG with a state probe: the factuality component

GROUNDHOG detects *grounding*—whether a response is supported by its context. A distinct question is *factuality*—whether it is true—which can come apart from grounding (a response can be grounded yet false). We ask whether adding a state-probing detector helps. We combine GROUNDHOG with a SAPLMA hidden-state probe (Azaria and Mitchell, 2023) through a learned two-parameter gate over their log-odds, $\sigma(a\, z_{\text{GROUNDHOG}} + b\, z_{\text{SAPLMA}} + c)$, fit and evaluated by nested cross-validation with no score normalization (Table 7).

On the grounding benchmarks the gate adds essentially nothing (± 0.004): GROUNDHOG already captures the grounding signal, and the state probe is both weaker and redundant with it. On HDM *factuality* (`is_hal`), however, the gate improves over GROUNDHOG by $+0.012$—a response's truth depends on semantic content in the accumulated state that a grounding detector does not read. The equal-weight sum underperforms both detectors, so this small gain requires learning the gate, not naive averaging. We report GROUNDHOG on its own as the grounding detector and note the state probe only as an optional factuality add-on.

	RAGTruth	RAGognize	RAGBench
GROUNDHOG	0.803	**0.854**	0.802
SAPLMA (state probe)	0.680	0.667	0.683
equal-weight logit sum	0.755	0.778	0.762
GROUNDHOG + SAPLMA (learned gate)	**0.804**	0.853	**0.803**

Table 7: **Combining GROUNDHOG with a SAPLMA state probe** (Llama-2-7B, 5-fold nested CV; GROUNDHOG at C=0.06). A learned two-parameter gate over the two detectors' log-odds, $\sigma(a z_{\text{GROUNDHOG}} + b z_{\text{SAPLMA}} + c)$ (no score normalization), adds essentially nothing on the grounding benchmarks ($\pm$0.004) but improves *factuality* detection by +0.012 (HDM `is_hal`), where a response can be grounded yet untrue. The equal-weight sum underperforms both detectors, so the small factuality gain requires learning the gate, not naive averaging. Best per column in bold.

Train ↓ / Test →	RAGTruth	RAGognize	RAGBench	HDM
RAGTruth	**0.803**	0.767	0.753	0.693
RAGognize	0.739	**0.854**	0.735	0.761
RAGBench	0.730	0.761	0.866	0.751
(training-free)	**0.803**	0.739	**0.877**	**0.773**

Train ↓ / Test →	RAGTruth	RAGognize	RAGBench	HDM
RAGTruth	**0.803**	0.756	0.815	0.691
RAGognize	0.802	**0.854**	0.815	0.721
HDM	0.683	0.705	0.850	**0.802**
(training-free)	**0.803**	0.604	0.824	0.682

Table 8: **Cross-dataset transfer of GROUNDHOG** (rr3, C=0.06; AUROC, Llama-2-7B). An ℓ_1 classifier is trained on the row dataset and evaluated on the column dataset (HDM = context-hallucination). Diagonal (**bold**) is within-dataset 5-fold CV; off-diagonal is zero-shot transfer. The bottom row is the training-free for reference: GROUNDHOG transferred from a *different* dataset still exceeds on every target, so the learned head captures domain-general grounding structure rather than dataset artifacts.

D Cross-Dataset Transfer

Does the supervised head learn transferable grounding structure, or dataset artifacts? We train GROUNDHOG on one dataset and evaluate it zero-shot on each of the others (Llama-2-7B), and contrast this with the *training-free* , which needs no data and is therefore the same detector everywhere. This puts three quantities on one axis (Figure 5): as a no-training floor, GROUNDHOG trained in-domain as the ceiling, and GROUNDHOG transferred zero-shot from another domain in between. The key observation is that *transferred* GROUNDHOG—trained on a different dataset entirely—still beats the training-free on every target (by +0.003 to +0.063): the learned head captures domain-general grounding structure, not dataset-specific shortcuts, while in-domain training adds a further increment. The full transfer matrix is in Table 8.

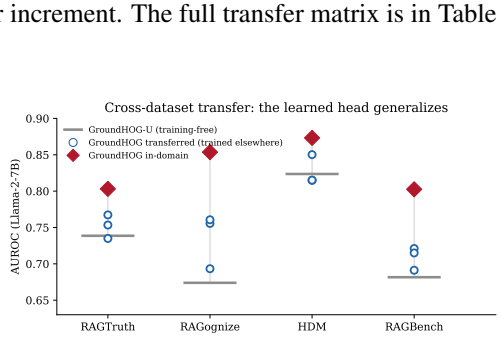


Figure 5: **Cross-dataset transfer vs. the training-free signal** (Llama-2-7B). For each target dataset: the training-free (grey), GROUNDHOG trained in-domain (red), and GROUNDHOG transferred zero-shot from each of the other datasets (open circles). Transferred GROUNDHOG sits above on every target—the supervised head generalizes across domains.

E Additional Benchmark Results

E.1 Newer and Larger Model Families

The main tables already span three families (Llama-2-7B, Mistral-7B, Qwen3-4B) as readers, with GROUNDHOG the top detector on 11 of 12 cells. To test whether the source-write signal extends to a contemporary open model and a newer, larger Qwen, we additionally evaluate **GPT-OSS-20B** and **Qwen3.5-9B** as proxy readers on the two proxy benchmarks (HDM, RAGBench), following the same extraction, preprocessing, and evaluation protocol (Table 9). GROUNDHOG remains the top detector on every model×benchmark cell—clearly ahead of the strongest steelmanned baseline and the training-free —with the largest margins on HDM. The signal is thus not an artifact of the model families used in the main evaluation; it recurs in a 20B open model and a larger Qwen release.

Reader model	Benchmark	GROUNDHOG	best steelman†
GPT-OSS-20B	RAGBench	**0.832**	0.813 (NoVo†)
	HDM	**0.884**	0.868 (Lookback†)
Qwen3.5-9B	RAGBench	**0.794**	0.789 (NoVo†)
	HDM	**0.887**	0.871 (Lookback†)

Table 9: **Newer and larger reader models.** GROUNDHOG (C=0.06), the strongest *steelmanned* baseline (†: NoVo or Lookback Lens features read through our ℓ_1 head, C=0.1), and the training-free , with GPT-OSS-20B and Qwen3.5-9B as proxy readers on the two proxy benchmarks (HDM context-hallucination, 5-fold CV; RAGBench, test split). GROUNDHOG is the top detector on every model×benchmark cell (by +0.005 to +0.019 over the best steelman). We also evaluated the SAPLMA hidden-state probe on these readers; it trails both steelmanned baselines on every cell (RAGBench ≈0.69, HDM ≤0.851) and is omitted. The source-write signal thus recurs in a contemporary open model and a larger, newer Qwen.

E.2 Self-Reader and Proxy-Reader Results

The main experiments read each response with the model that generated it (a *self*-reader). We test whether the source-write signal survives when a *different* model reads the same responses. On the RAGTruth responses generated by Mistral-7B, we extract features with the generator (Mistral, self) and with a proxy reader (Llama-2-7B), holding the examples, labels, split, pooling, and classifier fixed—only the reader model changes (Table 10).

GROUNDHOG transfers to the proxy reader at a cost of only 0.009 AUROC (0.873 → 0.864): a different model reading the same responses still recovers the signal. is essentially reader-invariant (0.808 → 0.808), as expected for a model-intrinsic logit attribution. The write-magnitude signal is thus not an artifact of a model reading its own generations.

Reader regime (RAGTruth, Mistral-generated text)	GROUNDHOG
Self (Mistral reads Mistral)	0.873
Proxy (Llama-2-7B reads Mistral)	0.864
	0.808
	0.808

Table 10: **Self- vs. proxy-reader.** Matched comparison: the same generated responses, read by the generator vs. a different model; identical examples, labels, split, pooling, and classifier. GROUNDHOG loses only 0.009 AUROC under the reader swap and is invariant.

E.3 Task- and Label-Level Breakdown

The aggregate AUROC numbers in the main tables average over heterogeneous strata: question types, answerability, domains, and different *notions* of hallucination. Here we disaggregate every benchmark to ask where the source-write signal is strong, where it is weak, and—crucially—whether its weak spots are specific to GROUNDHOG or shared by all detectors. Throughout we compare GROUNDHOG against the stronger of the two steelmanned baselines (NoVo and Lookback Lens, each fit through our own ℓ_1 head) and against the training-free .

Groundedness is readable; factuality and abstraction are not. Table 11 slices RAGTruth by task, RAGognize by answerability, and HDM by label type. The clearest pattern is the behaviour of the training-free : it tracks GROUNDHOG closely wherever the label is about *grounding in retrievable context* (RAGognize answerable 0.726, HDM grounding 0.824, RAGTruth QA 0.725), but collapses toward chance exactly where there is no context signal to read out. On *unanswerable* RAGognize questions falls to 0.554, on abstractive RAGTruth data-to-text (D2T) to 0.541, and on HDM *factuality* to 0.645. This is the expected failure mode: is a context-write logit attribution, so when a response cannot be checked against the source— because none is relevant (unanswerable) or the task is generative rather than extractive (D2T)—there is nothing for it to attribute. The supervised GROUNDHOG is far more robust in these strata (0.869, 0.771, 0.761 respectively) because it also reads the response-write magnitude, which remains informative when the context channel goes dark.

Benchmark	Stratum	n	pos.	GROUNDHOG	best steel†	
RAGTruth	QA	150	0.35	**0.732**	0.722 (NoVo)	0
	Summary	150	0.34	0.714	**0.760** (Look.)	0
	D2T	150	0.82	**0.771**	0.729 (Look.)	0
RAGognize	answerable	1391	0.41	**0.805**	0.788 (Look.)	0
	unanswerable	1390	0.69	0.869	**0.870** (Look.)	0
HDM	grounding	1118	0.75	**0.873**	0.863 (Look.)	0
	factuality	1118	0.45	**0.761**	0.749 (Look.)	0

Table 11: **Detector AUROC by task and label type** (Llama-2-7B). Each row is a stratum of one benchmark; *pos.* is the positive (hallucination) rate. GROUNDHOG is the response-anchored supervised detector, *best steel*† is the stronger of the two steelmanned baselines (NoVo / Lookback Lens, each fit through our ℓ_1 head), and is the training-free context-write signal. —Best of the three detectors per row in **bold**. Two patterns recur: (i) collapses toward chance exactly where there is no grounded content to read—*unanswerable* RAGognize questions (0.554), abstractive RAGTruth *D2T* (0.541), and *factuality* rather than grounding on HDM (0.645)—whereas GROUNDHOG stays strong; and (ii) factuality/reasoning strata are a blind spot for *all* detectors (HDM factuality ≤ 0.761 vs. grounding ≥ 0.824), confirming that these methods read *grounding*, not truth.

Factuality is a universal blind spot. The HDM rows isolate the two questions that grounding detectors conflate. On *grounding* (`is_ctx_hallucination`) every method scores well (≥ 0.824); on *factuality* (`is_hal`)—whether the statement is true, independent of the source—every method drops by roughly 0.11 (GROUNDHOG 0.761, best steelman 0.749, 0.645). The uniform drop confirms that these internal-state detectors, ours included, read whether a response is *supported by its context*, not whether it is *true*; a response can be faithfully grounded in an erroneous or incomplete source and still be false. This motivates the optional factuality add-on of §C.2, where a semantic state probe recovers part of the residual gap, and it delimits the scope of the paper's claims.

Difficulty is set by the domain, and it is shared. Figure 6 breaks RAGBench into its twelve source domains, sorted by GROUNDHOG AUROC, which ranges from 0.61 to 0.81. The hardest domains—financial-table QA (finQA 0.61), multi-hop reasoning (hotpotQA 0.64), and legal contracts (cuad 0.67)—share numeric, tabular, or multi-hop content, where a hallucination need not correspond to a shift in *which* source tokens are attended: the write pattern can look grounded while the arithmetic or the hop is wrong. This is the same grounding-vs-reasoning boundary seen on

HDM factuality, now resolved by domain. The difficulty is not specific to GROUNDHOG—the steelmanned baselines and bottom out in exactly these domains too, and GROUNDHOG is the best detector on 8 of 12 domains and on the pooled benchmark (0.795 vs. 0.775 for the best steelman), edged out only where all methods are near their floor and the ranking is within noise.

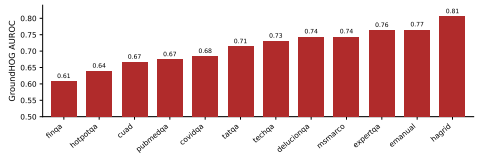


Figure 6: **GROUNDHOG across RAGBench source domains** (Llama-2-7B, AUROC), sorted. Difficulty is set by domain: reasoning-/table-heavy domains (finQA, hotpotQA, cuad) are hard, extractive domains (emanual, hagrid) are easy; the same ordering holds for the baselines (§text).

F Response length: confound analysis and controls

Response length is correlated with hallucination in every benchmark we use (length alone predicts the response-level label at AUROC 0.74/0.60/0.66 on HDM/RAGTruth/RAGognize). Because our per-head write magnitudes are *token means*, they inherit a strong length dependence, and the two streams inherit it with *opposite* sign: as a response grows, later tokens attend more to the response history and less to the context, so $\|w_r\|$ rises with length and $\|w_c\|$ falls (Table 12). This appendix documents the confound, shows that the supervised detector is robust to it, and explains why we present the mechanism at the token level (§**??**) rather than from the response-level statistic.

Table 12: Length correlations. $\rho(\text{len}, \cdot)$ is Pearson correlation of pooled log-magnitude with log response length. Both streams are heavily length-dependent, oppositely.

	$\rho(\text{len}, \text{label})$	$\rho(\text{len}, \|w_r\|)$	$\rho(\text{len}, \|w_c\|)$
HDM	+0.36	+0.74	−0.73
RAGTruth	+0.17	+0.90	−0.84
RAGognize	+0.29	+0.92	−0.88

Confound or mediator? Length is partly a genuine mediator—the hallucination process tends to produce both longer generations and more self-sourced writing—and partly a true confound, because the response-level label is positive if *any* span is unfaithful, so longer responses are mechanically more likely to be labeled hallucinated (a larger error surface). Two observations bear on this. First, controlling for the writing signal collapses length's predictive power (e.g. on RAGognize, AUROC(len) falls from 0.66 to 0.51 once residualized on $\|w_r\|$), whereas controlling for length leaves the writing signal a clear edge ($\|w_r\|$ retains 0.57–0.69): the write subsumes length more than the reverse. Second, length still predicts hallucination *within* each RAGTruth task type (AUROC 0.65/0.64/0.54 on D2T/QA/Summary), so it is not merely a proxy for task. We therefore treat length as a mixed effect and do not residualize it out of the detector; instead we report the length-robust operating point directly.

The supervised detector is length-robust. Table 13 reports GROUNDHOG with three controls: raw, linearly residualized on log-length (an over-control, since length is partly a mediator), and averaged *within* length quartiles. The within-quartile numbers—the fair control—stay at 0.79–0.84, well above length alone and far above the pooled single-stream residuals. The ℓ_1 head selection therefore exploits length-independent per-head structure; the method is not a length detector.

Table 13: GROUNDHOG under length control. Within-quartile AUROC (holding length fixed) is the fair control; it stays high, confirming length-independent signal.

	raw	length-residualized	within-length-quartile
HDM	0.872	0.809	0.837
RAGTruth	0.798	0.779	0.786
RAGognize	0.847	0.801	0.823

Why the mechanism is reported at the token level. The *response-level* version of the opposite-pole statistic is large but not trustworthy: the per-head sign agreement (coherence ≈ 0.9, with 99% of top response heads and 1% of top context heads pointing toward hallucination) is manufactured by the shared length dependence—every head co-moves with length, and length co-moves with the label. Residualizing or binning on length collapses that agreement (coherence to ≈ 0.1–0.5; the near-unanimous split degrades to roughly 20%/77%, and on RAGTruth the response pole flips). The sign coherence is thus a statement about aggregate length co-movement, not head-level computation. The within-response, token-level analysis in §**??** avoids this entirely—length is held fixed and each token carries its own label—and the split survives there, which is why we make that the primary evidence.

F.1 Two-Sided Attribution

Answer-conditioned attribution identifies the sources whose states are read when the answer token is written. In multi-hop questions, some supporting sources contribute mainly indirectly, for example by establishing an intermediate entity or relation that later computation depends on. This type of contribution is not included in the direct term. As an approximation, the same decomposition is applied a second time with the

question tokens as targets. This requires that the question follows the sources in the input. The two scores are

$$g_k^A = \mathrm{GH}(S_k \mid y), \qquad g_k^Q = \mathrm{GH}(S_k \mid q). \tag{25}$$

The two-sided ranking first selects the source with the highest answer-conditioned score,

$$k^\star = \arg\max_k \; g_k^A, \tag{26}$$

and then ranks the remaining sources by g_k^Q. The first score identifies the source that directly supports the answer, and the second identifies sources that are read in relation to the question. The same attribution rule is used for both scores, and no parameters are learned to combine them.

G Two-Sided Attribution

Answer-conditioned attribution naturally emphasizes evidence that directly writes the answer. In multi-hop questions, however, a complementary source may primarily establish an entity or relation expressed in the question rather than the final answer. We therefore apply the same decomposition a second time with the question tokens as targets:

$$g_k^A = \mathrm{GH}(S_k \mid y), \qquad g_k^Q = \mathrm{GH}(S_k \mid q). \tag{27}$$

Our two-sided ranking first selects the answer-conditioned anchor

$$k^\star = \arg\max_k g_k^A, \tag{28}$$

then ranks the remaining sources by g_k^Q. The first side therefore identifies evidence that writes the answer, while the second recovers complementary evidence that explains the query. This variant uses the same attribution rule in both passes and adds no learned fusion parameters.

H Additional Qualitative Examples

The main combined attribution heatmap is discussed in §**??**; additional token-level examples across RAGTruth cases are collected in Figure 7.

I Computational and Implementation Details

Table 14 reports per-response latency by cost class.

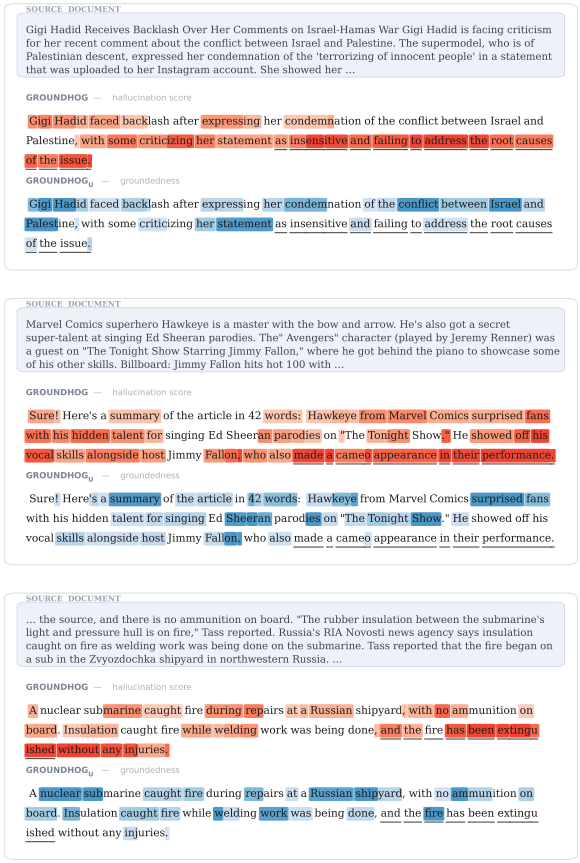


Figure 7: **Additional token-level attribution examples** (RAGTruth). In each example, the source document is shown once, and the response is shown twice: coloured by GROUNDHOG's per-token hallucination score (red) and by 's per-token groundedness (blue). Hallucinated spans are underlined.

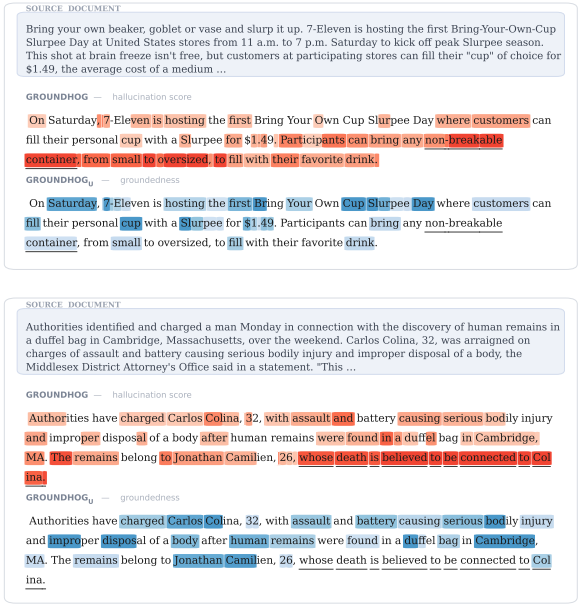


Figure 7: (continued) Additional token-level attribution examples (RAGTruth).

Method	Access	Fwd. passes	Gen. (K)	Judge calls	Lat. (s) /resp.
	unsup, white-box	1	–	–	0.08
GROUNDHOG	sup, white-box	1	–	–	<1
ReDeEP / NoVo / Lookback / TPA / SAPLMA	sup, white-box	1	–	–	<1
Perplexity / Entropy	unsup, white-box	1	–	–	0.08
LUMINA	unsup, white-box	1	–	–	<1
NLI	unsup, black-box	–	–	n_{claim}	judge
RAGAS	unsup, black-box	–	–	$1+n_{\text{stmt}}$	judge
INSIDE	unsup, white-box	10	10	–	9.2
SelfCheckGPT	unsup, black-box	–	10	$K \cdot n_{\text{sent}}$	9.2+judge

Table 14: **Inference cost per response** (Llama2-7B, NVIDIA L40S). White-box detectors share a single forward pass over the (context, response) pair ($\approx$ 78 ms); GROUNDHOG adds per-head write magnitudes and an ℓ_1 logistic, staying well under a second. INSIDE and SelfCheckGPT instead require $K=10$ stochastic generations (≈ 9.2 s, a $\sim 118\times$ gap), and SelfCheckGPT/RAGAS/NLI additionally invoke a separate judge LLM ($n_{\text{claim}}/n_{\text{stmt}}/K \cdot n_{\text{sent}}$ calls). is the cheapest: one forward pass, no training, no judge.

AI use statement

AI tools were used for some aspects of this project.

Tasks with required disclosure.

Tasks with recommended disclosure. We have reviewed all AI-assisted work described above. Any errors are the authors' own.